IEEE Access
Multidisciplinary | Rapid Review | Open Access Journal

SPECIAL SECTION ON AI AND IOT CONVERGENCE FOR SMART HEALTH

Received April 22, 2021, accepted May 4, 2021, date of publication May 7, 2021, date of current version May 18, 2021.

*Digital Object Identifier 10.1109/ACCESS.2021.3078241*

# Deep Learning-Based Smart IoT Health System for Blindness Detection Using Retina Images

**AMIT KUMAR JAISWAL[1], PRAYAG TIWARI[2], SACHIN KUMAR[3], (Senior Member, IEEE), MABROOK S. AL-RAKHAMI[4], (Member, IEEE), MUBARAK ALRASHOUD[5], (Member, IEEE), AND AHMED GHONEIM[5,6], (Senior Member, IEEE)**

[1]School of Mathematics, University of Leeds, Leeds LS2 9JT, U.K.
[2]Department of Computer Science, Aalto University, 02150 Espoo, Finland
[3]Department of System Programming, South Ural State University, 454080 Chelyabinsk, Russia
[4]Information Systems Department, College of Computer and Information Sciences, King Saud University, Riyadh 11543, Saudi Arabia
[5]Department of Software Engineering, College of Computer and Information Sciences, King Saud University, Riyadh 11543, Saudi Arabia
[6]Department of Mathematics and Computer Science, Faculty of Science, Menoufia University, Shebin El-Koom 32511, Egypt

Corresponding author: Ahmed Ghoneim (ghoneim@ksu.edu.sa)

This work was supported by the Deanship of Scientific Research at King Saud University, Riyadh, Saudi Arabia, through the Research Group under Project RGP-229.

**ABSTRACT** Deep Learning-based Smart Healthcare is getting so much attention due to real-time applicability in everyone life's, and It has obtained more attention with the convergence of IoT. Diabetic eye disease is the primary cause of blindness between working aged peoples. The major populated Asian countries such as India and China presently account for millions of people and at the verge of an eruption of diabetic inhabitants. These growing number of diabetic patients posed a major challenge among trained doctors to provide medical screening and diagnosis. Our goal is to leverage the deep learning techniques to automate the detection of blind spot in an eye and identify how severe the stage may be. In this paper, we propose an optimized technique on top of recently released pre-trained EfficientNet models for blindness identification in retinal images along with a comparative analysis among various other neural network models. Our fine-tuned EfficientNet-B5 based model evaluation follows the benchmark dataset of retina images captured using fundus photography during varied imaging stages and outperforms CNN and ResNet50 models.

**INDEX TERMS** Diabetic retinopathy, medical diagnosis, CNN, retina images, the IoT.

## I. INTRODUCTION

In recent years, AI and IoT convergence for efficient smart healthcare systems is getting attention in the community [1]–[4]. This convergence allows detecting different health disease more efficiently than ever before. Diabetes is one of the chronic diseases around the world that occurs when our body is not able to produce insulin hormone or not able to use it effectively. World Health Organization (WHO) reported that diabetes caused more than 1.6 million deaths in 2016 [5].Diabetic patients tend to produce a high glucose level in the blood, which may cause damage and failure to the body's organs. According to the International Diabetes Federation (IDF), 1 out of 10 people is diabetic, which is a serious concern. Possible complications of diabetes may result in heart stroke, kidney failure, serious problems with eye vision etc.

The associate editor coordinating the review of this manuscript and approving it for publication was Abdulsalam Yassine.

One of the serious complications of diabetes is Diabetic Retinopathy (DR). DR may cause complete blindness and many people around the world are victims of DR [6], [7]. Around 25% of people out of total diabetic patients are affected by DR only, which makes DR a complicated disease [8]. Long-term diabetes may result in DR, which is a progressive disease and may cause partial or permanent vision impairment. Majority of people affected from DR are working-age group, which are the main workforce of the growing economy of any country [9]). IDF also reported that only India alone shares a large portion of diabetic people around the world and this share is growing rapidly every year [9].

It is difficult to detect the symptoms of DR in the early stage, which is a challenging and important issue in medical science. The initial symptoms of DR are very light and patients are usually unaware until it results in irreversible damage to the retina or it is diagnosed through a medical test. Therefore, it is highly desirable that DR must be detected


This work is licensed under a Creative Commons Attribution 4.0 License. For more information, see https://creativecommons.org/licenses/by/4.0/

at the earliest. However, it is possible to detect DR with the availability of highly efficient and trained person who can evaluate the digital color fundus photographs of the retina. The rear part of the human eye is known as fundus. These fundus images are evaluated by locating the lesions associated with vascular abnormalities that arise due to diabetes. Therefore, the effective solution is available but it is a time-consuming process and demands the availability of the highly efficient medical practitioner.

Deep learning is known to be a very popular approach in different healthcare, patient monitoring and the systems, which requires huge medical image processing [10]–[16]. Convolutional neural network (ConvNets) is a deep learning approach, which is very effective for image data analysis in several domains [13]. he accuracy of ConvNets can be improved by scaling up different parameters subject to the availability of resources. These scaled versions of ConvNets are very efficient for the medical domain where accuracy is highly desirable. Therefore, in this study, EfficientNet architecture [17] is used to analyze the retina images in order to detect the DR.

### A. CONTRIBUTION

The major contribution of this work is as follows:

- We utilized state-of-the-art EfficientNet model for identifying blindness symptoms at the first instance and found significant improvement to identification of blindness in retinal images with over 92% validation accuracy outperforming CNN and ResNet50 models.
- We proposed a novel augmentation step called as *Polar unrolling* for this imbalanced dataset of retinal images which significantly improves the prediction accuracy during test time augmentation.

### B. ORGANIZATION

The rest of the article is organized as follows: Section 2 provides state of art literature review. In Section 3, the methodology used is discussed in detail. Section 4 provides the details about experiments followed by the evaluation and discussion of the results in Section 5. The article concludes with a discussion of future scope in section 6.

## II. RELATED WORK

Researchers [18]–[23] have been working in the area of connected smart health [22], healthcare Internet of Things (HealthIoT) [24], patient monitoring [25], where AI and IoT technologies have considerable potential. AI tools and technologies, particularly for Diabetic Retinopathy, will enhance the state of global health. Kumar and Vashist [26] discussed the various challenges and achievements in eye care with a focus on Indian community. They mentioned that several eye diseases that may results in vision loss should be provided effective care and potential soultions by 2020 in India. Bhalla *et al.* [27] proposed a model to demonstrate an innovative modality to address the issue of DR in India by organizing a certification programs for technicians and doctors to improve the knowledge and skills required for DR detection at early stages.

Diabetic Retinopathy is a very common disease among diabetic patients that can lead to partial or complete blindness. Therefore, early diagnosis and detection are very important. The existing state of the art diagnosis methods to detect DR requires continuous observation of patients by a skilled physician. These diagnosis methods are time-consuming and subject to the availability of expensive medical equipment.

In [18], the researchers mentioned that only a group of 10-15 physicians is responsible for manually diagnosing DR over 2 million retinal images per year in the largest eye care facility in the world named as Aravind Eye Hospital, India. They have reinstated the extreme effort in terms of infrastructure and time required in the task when a large number of cases turn out to be normal.

Automatization of this process can assist the doctors in diagnosing the DR patients effectively and to save the time spent on diagnosing the normal cases. Considering this aspect in mind, several researchers have worked in this area and utilized the machine learning artifact to devise a model that can predict the presence of DR in a patient. In the following paragraphs, we review some of these key researches in this field.

In [18], they presented an account of the initial efforts they put into the development of an automated system using computer vision techniques to operate patients in the early stage of DR using retinal color fundus images. The project was a continuation of their previous work on WiLDNet at Aravind eye hospital. Their main focus was to create a model based on different retinopathy types using SIFT descriptors. The important features include hemorrhages and exudates. The intention was to develop a robust and long-term model. Besides that, Support Vector Machine classifier is trained to label each image patch; post, which these images are aggregated and a decision over the entire image is identified, based on patch-level prediction. The authors' reported an equal error rate of 87% using 1000 images.

In [20], the researchers developed a solution for early detection of DR using deep convolutional neural network technique to detect micro-aneurysms (MAs) in diabetic patients. They also performed multi-label classification [28] by assigning labels to retinal fundal images on five categories. Earlier they proposed multi-layer convolutional neural networks (CNNs) by considering two fully connected layers and a single output layer to efficiently detect DR. Catering to the issue of oversampled classes, they use a small capacity network with L2 regularization and dropout, $4 \times 4$ kernel, pre-processing and augmentation methods. The performance of the approach was evaluated on Kaggle diabetic retinopathy dataset. dataset.[1] The approach gave a kappa score of more than 0.85 in severity grading. The approach was claimed to be efficient in both time and space complexity. However,

[1] https://www.kaggle.com/c/diabetic-retinopathy-detection/data

a mean squared error is used as an objective function on the regression problem.

A model built using convolutional neural networks such as AlexNet, VggNet, GoogleNet and ResNet is proposed in [23] to identify Diabetic Retinopathy in diabetic patients. The fundus image of DR are classified into five classes based on which the model identifies the stage of DR for the patient. Moreover, the use of transfer learning and hyper-parameter tuning leveraged the CNNs model for better accuracy [29], which was not possible before using non-transferred learning for noisy data. Normalization and data augmentation techniques have been deployed for preprocessing of images and non-local means denoising (NLMD) methods were deployed to eject noise. The performance of the system was tested and evaluated on datasets available on Kaggle and classification accuracy of 95.68% was reported to the application of CNN and transfer learning.

An EyeWeS model was proposed to achieve high performance and greater efficiency by converting the pre-trained convolutional neural network architecture for DR detection into a weekly-supervised model by eliminating the data automation lesion-wise required for pixel-level training [30], [31]. Moreover, the researchers not only focused on the identification of diabetic retinopathy but also directed to the region of the eye that got effected. Besides, this model consists of bag labels to train instances (i.e, image patches) to identify the pooling function $f$ to encode relation among instances and bag labels. The objective of pooling function is to train instance classifier $P(y_i|x_i, \theta)$. They have checked the performance of their scheme on the Messidor dataset. On E-optha MA, micro-aneurysms of size 3 pixels in diameter could be detected with a significant improvement in the results has been obtained from Inception V3 with a higher AUC of 95.8% using only 5% of the Inception V3's number of parameters. A remarkable 97.1% AUC (area under curve) has been reported for a cross-dataset experiment.

Past work [32] employed a deep learning methodology to detect the diabetic retinopathy (DR). They gave attention to regression activation Maps (RAM) model to capture the discriminative area of input retina image. The network architecture was proposed by authors' considered that the global averaging pooling (GAP) layer to identify the total contribution in final prediction mentioned by neuron. The RAM model is depicted by $G(i,j) = \sum_{k=1}^{K} g_k(i,j)w_k$. In addition, the comparison of their own neural network with conventional neural network such as AlexNet, GoogLeNet is highlighted in their work. The authors' proposition claims better performance due to the addition of RAM that distinguishes the exact area of severity in terms of vascular abnormalities.

## III. METHODOLOGY

The ConvNets can be scaled based on the requirements to achieve better accuracy subject to the availability of resources. There are several ways to achieve this scalability. For example, we can increase the number of layers in ResNet [33] to scale it up from ResNet-18 to ResNet-200. The popular approaches to scale up a ConvNets model is to increase the width or depth of the model or a high resolution image can be used for model training and testing. However, these approaches do not use a well defined criteria to select the width, depth or resolution of the input image. In this study, EfficientNet [17], a balanced and more accurate deep ConvNets is used to detect DR in retina image data set which uses a well defined criteria to scale up on all dimensions and provides better accuracy.

### A. MODEL

This section details the adapted recent convolutional neural network-based EfficientNet model [17] for Diabetic Retinopathy – a disease caused by damage to the retina. If not treated, it can progress to blindness. Existing method of diagnosis is defined by classes 0 to 4, where 0 represents no presence of diseases and 4 represents severe progression. Our proposed model is based on the EfficientNet-B5 [17] with ImageNet as pre-trained weights, and also with different input image sizes of $224 \times 224$ and $256 \times 256$ respectively. The EfficientNet uses a compound dimension scaling approach [17]. This approach uses a coefficient parameter $\phi$ to apply uniform scaling on width, depth and resolution as given in equation 1 as follows:

$$\begin{aligned} &width : w = \beta^{\phi}, depth : d = \alpha^{\phi}, resolution : r = \gamma^{\phi}, \\ &\text{such that } \alpha \cdot \beta^2 \cdot \gamma^2 \approx 2 \\ &\alpha \geq 1, \quad \beta \geq 1, \ \gamma \geq 1 \end{aligned} \tag{1}$$

$\phi$ is a user defined coefficient and acts as a control parameters to measure the availability of resources for model scaling. $\alpha$, $\beta$ and $\gamma$ are constants to control the number of resources to be assigned to scale width, depth and resolution. In the main architecture of EfficientNet-B5, we replaced the top head with GlobalAveragePooling2D and a dense layer as linear output. Our models uses mean squared error as a loss function and the model output were linear activation function.

Our aim is to optimize the quadratic weighted kappa score [34], [35] which we consider as a regression problem. In order to be flexible in the optimization and we can yield higher scores than solely optimizing for accuracy. We optimize a pre-trained EfficientNet-B5 with a few added layers. The metric that we optimize is the mean squared error which is the squared mean differences between the predictions and labels, as reported in the formula below. By optimizing the given metric we are also optimizing for quadratic weighted kappa in case if we round the predictions afterwards.

$$\frac{1}{n}\sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2 \tag{2}$$

where n is the number of data points, $Y_i$ denotes observed values, and $\hat{Y}_i$ denotes predicted values.

Since we do not have much training data (3662 images), we augment the data to increase the robustness of our proposed model. We also rotate the data on any angle. Also,

we flip the data both horizontally and vertically. Finally, we divide the data by 128 for normalization.

Also, we found one of the earlier work [36] which developed a technique to deal with imbalanced image data (in our study, its' retinal images) by sampling method, but it does not improved our prediction due to augmentation step as we remove black background, without augmentation it improves a little which our model seems to overfit due to continuous loss at the same rate without learning any training parameters even on using cyclic learning rate scheduler.

## IV. EXPERIMENTAL EVALUATION

In this section, we will detail the employed dataset followed by the pre-processing and data augmentation techniques as well as the training steps including inference of trained models.

### A. DATASET

In this work, we employ a real-world eye disease image dataset which has a large collection of high resolution retina images captured using fundus photography[2] underlying varied circumstances of imaging situations. A trained clinician has assessed the existence of diabetic retinopathy in each image on standard ICDR (International Clinical Diabetic Retinopathy) severity scale of 0 to 4. The rating scores of each DR causes are $\{0-No\ DR\}$, $\{1-Mild\}$, $\{2-Moderate\}$, $\{3-Severe\}$, $\{4-Proliferative\ DR\}$. The collection consists of 5,590 images labeled with a subject for each eye image out of which 3,662 and 1928 images are splitted as train and test set with corresponding classes of 2 and 1, respectively.

The size of images are quite large which is 2,896 × 1,944. According to Figure 1, it shows that the training data is highly unbalanced due to larger shift in "No DR" class as compared to "Severe" is very less. The dataset is noisy in terms of images and labels such as some of the image artifacts are out of focus, overexposed or underexposed which is due to collection from various clinics using different camera over a long period of time.

[2]https://en.wikipedia.org/wiki/Fundus_photography

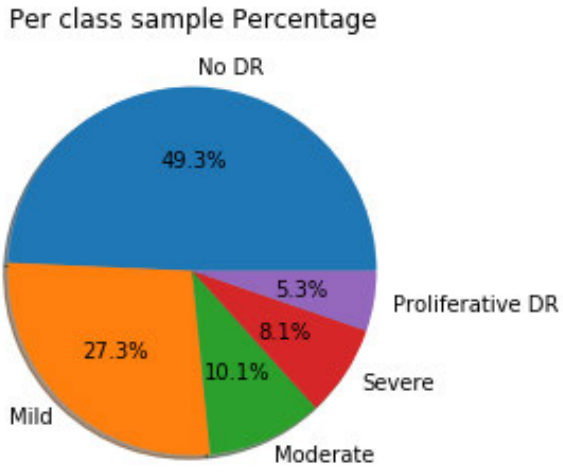



**FIGURE 1.** Class distribution.

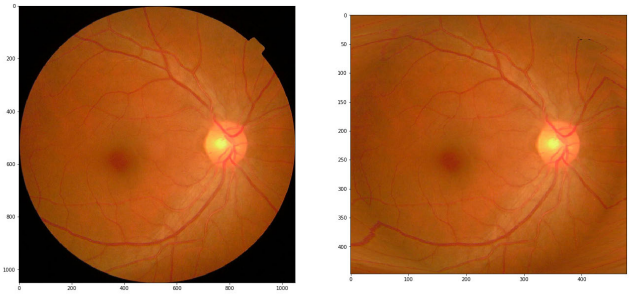


**FIGURE 2.** Typical augmentation-resize and crop.

### B. PREPROCESSING

This section details the data processing step which we performed before actually modeling this regression task so that training can be made faster on such unbalanced dataset. First and foremost, we transformed each retinal images in RGB channels followed by circular crop on each images based on image center to separate black regions from real-squared images by keeping the original aspect ratio so the images appearance natural. Also, we employ Ben Graham's method[3] to improve the illumination condition of images so that we can enrich the insights from eye images. Then, we resize the original images to $224 \times 224$ and $256 \times 256$ (different sizes to check the correctness / feasibility of our modeling approach), and we trained these collection of reduced image sizes with a large batch size of 32.

After preprocessing the dataset, we found that images when zoomed to center are exhibiting close relation to the light spots on left. To solve this zoom issue as reported in Figure 3, we performed augmentation on these processed images to make sure it generalises better and does not overfits.

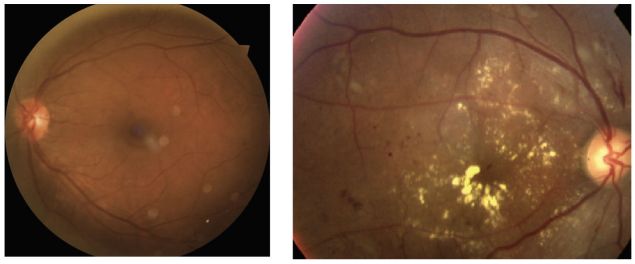


**FIGURE 3.** Preprocessed sample image: Left image is zoomed on the right.

### C. DATA AUGMENTATION

For better generalisation of the processed dataset, we look into this problem by executing some augmentation steps for

[3]https://bit.ly/2kTdTVe

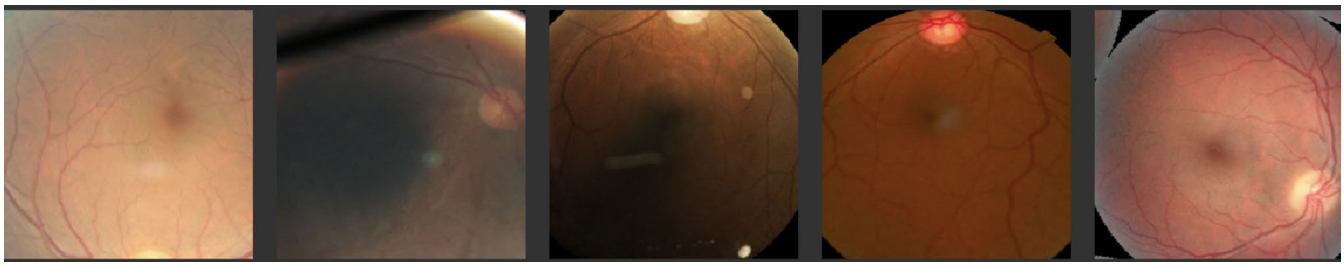


**FIGURE 4.** Sample images after augmentation.

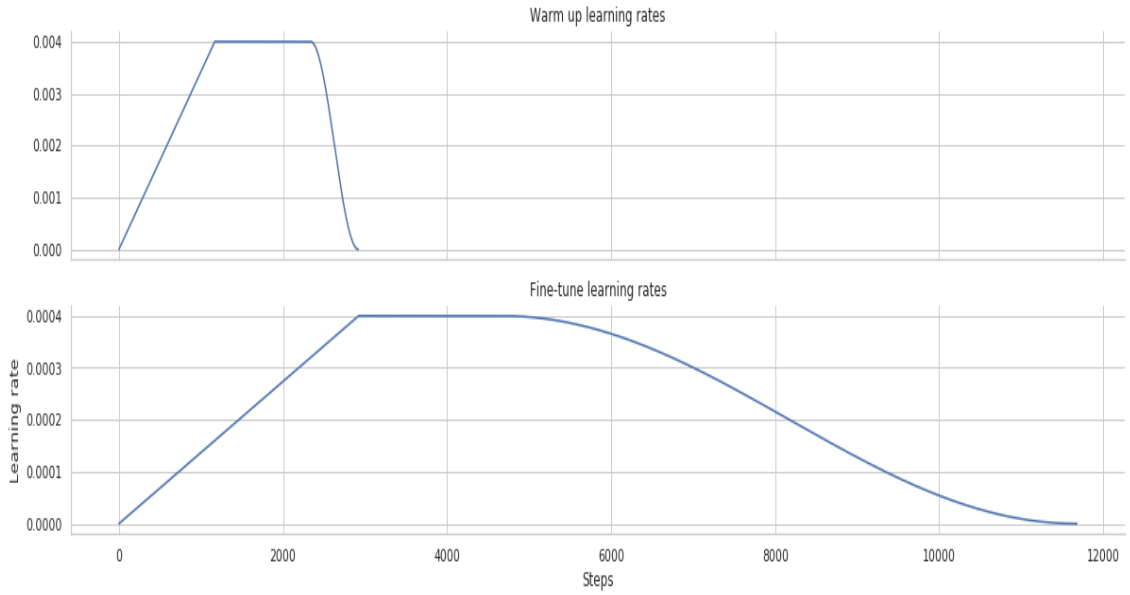



**FIGURE 5.** Learning rate policy scheduler.

the images. We employ the Albumentations library [37] to perform augmentation, where we flipped images horizontally and vertically, image rotation to 360°, zoom images to 1.3$x$ and contrasting for lightning. We report these augmentation steps in Figure 4.

We adapted a new augmentation step i.e. *Polar unrolling* which allows to better leverage pixel space, separate ''rotation'' from an augmentations list and acquire uniformly scaled eye images (for the instances of no or partial cropping of the fundus image with preservation of radius).

Originally we have images with noticeable black regions. In order to whittle away such regions we at the beginning apply an autocrop. After autocrop, we have the circle's radius (depicted by the broadest side of an image). Then, extraction of circle using polar unrolling. By unrolling, we changed coordinate space which do not require rotation augmentation. As rotation becomes just plain shift by x-axis (which does not matter for convolutional neural networks). It is more than absence of this type of augmentation. This made all possible rotations for our model considered (except some borders, which can be tackled by the single 50% shift by x-axis).

Our augmentation technique improves the existing contour to transform method on retinal images [6] in terms of processing the imbalanced data and training technique on the state-of-the-art EfficientNet models [17].

Apart from the above augmentation approaches, we also look into a specific oversampling technique [38] to compare the methodological flaw if there any exists in Polar unrolling as key part of the augmentation step. The oversampling of the data can be an alternative step to augmentation due to given imbalanced data. We used image size of 224 $\times$ 224, where the oversampling is performed after partitioning the training dataset as opposed to the steps considered in [38]. We found the oversampling approach [38] containing large number of duplicates images i.e., 22.45% of training set are not accounting for duplicates. In our oversampling strategy, we found no duplicate images and so a proper oversampling of the retina images dataset. Also, we used Mixup [39], a data augmentation technique along with oversampling for better generalisation of the detection result. The result of oversampling on the complete dataset is shown in Figure 8, where in the first row - the confusion matrix represents the oversampling before data partitioning and after partitioning

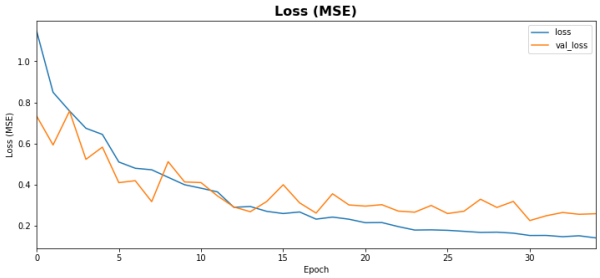



**FIGURE 6.** Training loss.

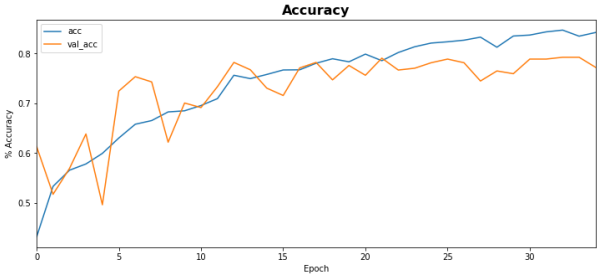



**FIGURE 7.** Accuracy.

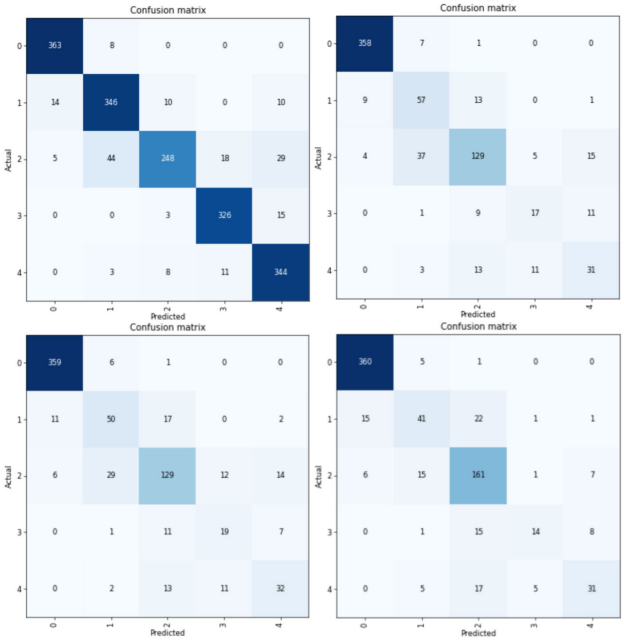



**FIGURE 8.** Oversampling result.

(in the first row and second column). In the second row of Figure 8, the first confusion matrix uses Mixup augmentation with oversampling, whereas the second confusion matrix is a result of Mixup augmentation [39].

### D. TRAINING AND INFERENCE

We trained our proposed EfficientNet-B5 based model with a larger batch size of 32 for 5 epochs as in the warm-up stage where we freeze all layers except the last two given the learning rate of *4e-3*, and we use Adam optimizer and cosine with learning rate (LR) scheduler. In the complete model training, we trained over 30 epoch under the fine-tuning stage provided all layers are unfreeze. Also, we use Early stopping monitoring validation loss for 5 epoch. The LR scheduler at different steps of training is schematically shown in Figure 5 followed by the training and validation loss in Figure 6.

As we found that the training loss becomes smoother around $\sim$ 0.4 and steeply improved to less than 0.2 at 30th epoch. This shows that TTA keeps the retinal feature during validation while training and could be improved if trained for long on higher batch size.

We cross-validated our EfficientNet-B5 model 5 folds and then inference these trained models by doing test-time augmentation (TTA) [40] 10 times and taken average of 5 models.

We employ 1xP100 and 1xT4 NVIDIA Tesla GPU for training models.

### E. EVALUATION MEASURES

To evaluate the detection model, we use *quadratic weighted kappa* (or Cohen' s Kappa), which measures the consensus between expert ratings and submitted ratings. The quadratic weights triggers an optimization factor in the rounding operation. This metric ranges from 0 (random agreement among raters) to 1 (complete agreement among raters). A perfect score of 1.0 is allowed when both the actuals and predictions are same, otherwise the least possible score is -1 which is provided when the predictions are furthest away from actuals. In this work, we treat all actuals as 0's and all predictions as 4's. This will give rise a quadratic weighted kappa score of $-1$.[4] The weighted kappa is given below:

$$\kappa = 1 - \frac{\sum_{i=1}^{k}\sum_{j=1}^{k} w_{ij}x_{ij}}{\sum_{i=1}^{k}\sum_{j=1}^{k} w_{ij}m_{ij}} \tag{3}$$

where we are going to optimize mean squared error (MSE) and by optimizing MSE, we will also optimize quadratic weighted kappa as the problem is expressed as regression.

## V. EXPERIMENTAL RESULTS

This section reports the evaluation result which is the predicted labels of DR for blind spot. For evaluation we predict values from the generator and round it to the nearest integer to get valid predictions. After that we compute the quadratic weighted kappa score on the training set and the validation set which is reported in Table 1.

We perform Grid search in order to optimize the validation score over a range of thresholds and the quadratic weighted

[4]Generally, the score near to 1 is best

**TABLE 1.** Cohen kappa score-fine-tuned efficientNet-B5 regression.

| Training Set | Validation Set |
|---|---|
| **0.96754** | **0.92321** |

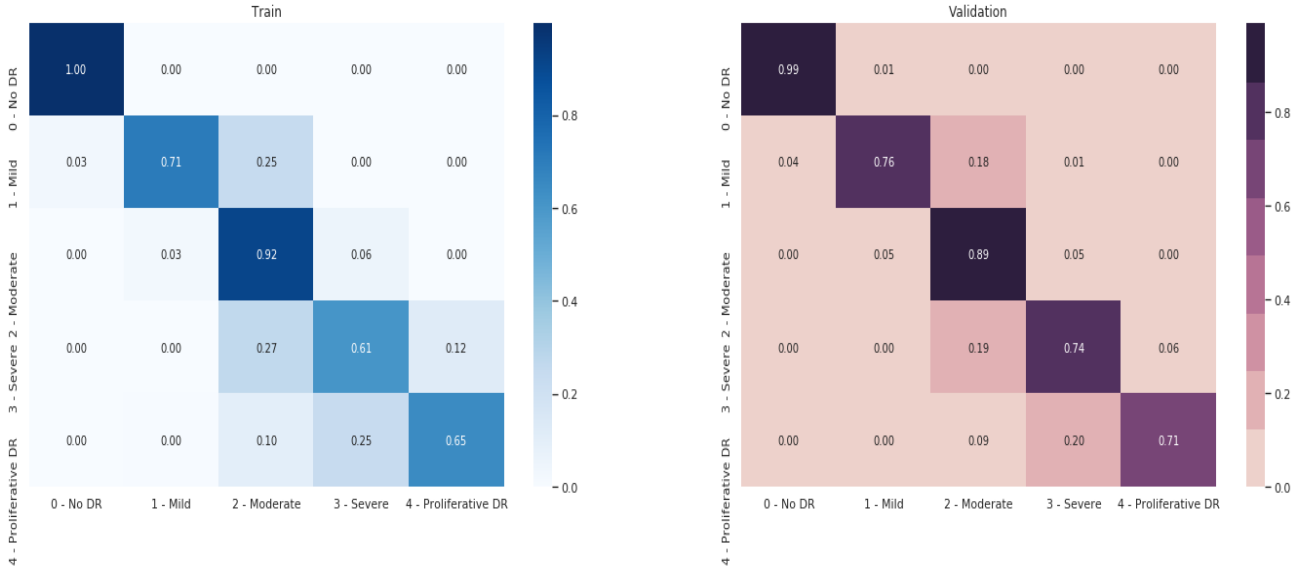



**FIGURE 9.** Confusion matrix-inference of 5 folds.

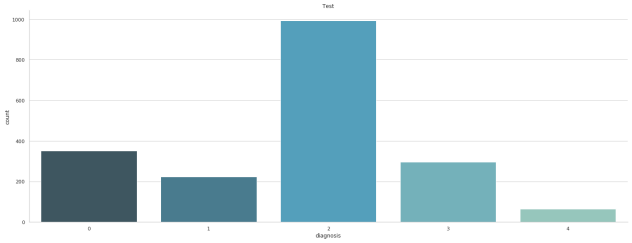



**FIGURE 10.** Label distribution-prediction result.

**TABLE 2.** Comparison of CNN models with efficientNet weights.

| Model | QWK Score |
|---|---|
| EfficientNet-B0 | 0.883 |
| EfficientNet-B1 | 0.892 |
| EfficientNet-B2 | 0.893 |
| EfficientNet-B3 | 0.900 |
| EfficientNet-B4 | 0.891 |
| **EfficientNet-B5** | **0.902** |
| ResNet50 | 0.858 |
| InceptionNetResNetV2 | 0.884 |
| DenseNet169 | 0.881 |

kappa score is 0.92 with threshold range of (0.5, 1.5, 2.5, 3.5). Our fine-tuned EfficientNet-B5 based model scores 92.32% on validation set with the input image size of $224 \times 224$.

We perform test time augmentation (same augmentation steps as mentioned in above section) 10 times and averaged all 5 models with their TTA predictions. The prediction result is reported in Figure 8 and Figure 9.

We adopted several convolutional neural network based models reported in Table 2 and found that DenseNet169 perform significantly better than ResNet 50 which is since former model improves on Adam optimizer.

The prediction scores of convolutional neural network models reported in Table 2 treats the identification problem as regression which we also considered for our proposed model on top of EfficientNet-B5 model. The models are trained in general without any fine-tuning steps and EfficientNet-B5 performs better than any other CNN models with a significant improvement of 0.10%.

## VI. CASE STUDY OF SMART IoT BASED HEALTHCARE SYSTEM

IoT based healthcare systems are more powerful by integration of deep learning approaches. Recent advancements in IoT technology have revolutionized the electronic healthcare research and industry applications. The huge increase in the use of portable smart health devices has increased the quality of health monitoring, diagnosis and collecting data for the clinicians with potential to perform early diagnosis and provide necessary treatment on time. However, the use of personal medical data and records is a concern of data security and data sharing policy [47]. Blockchain [41] provides a solution to deal with privacy and transparency of data. The typical IoT blockchain platform for smart healthcare is illustrated in Fig 10.

The framework consists of vital sign monitoring system, an IoT server [42], a blockchain network, and a communication interface to collect the patients information from the healthcare sensors. All the information is stored securely and communicated to the medical staff for further diagnosis and treatment. This information further can be used to develop decision making models using deep learning models to provide accurate diagnosis in time. The approach once development can be optimized to develop an IoT based smart device and can be used by the medical staff efficiently.

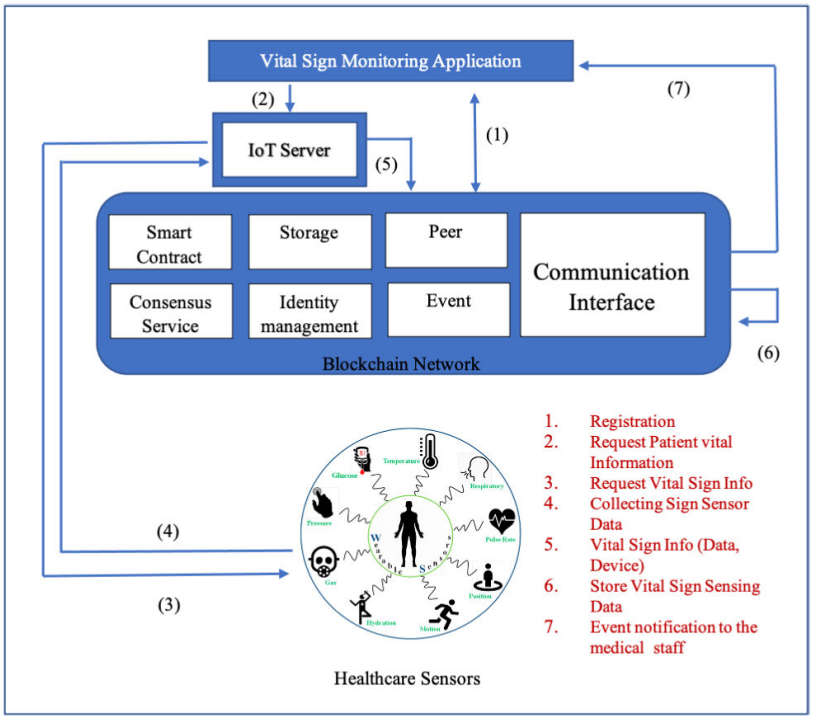



**FIGURE 11.** IoT based framework for smart healthcare.

## VII. CONCLUSION AND FUTURE WORK

In this work, we introduce the state-of-the-art DL based smart health system for identification of blindness in eye disease (diabetic retinopathy) evaluated on a retinal image dataset in the IoT. We have shown that the convergence of IoT with AI can lead to provide effective smart health system.

Our fine-tuned Efficient-B5 based model outperforms CNN and ResNet50 models with 92.32% validation accuracy which predicts diagnosis of diabetic retinopathy severity (eye blindness) on the five-point scale from retinal images. Our baseline model Efficient-B5 (with fine-tuning) model training on the average doctor opinion, a tactics that output state-of-the-art results on identifying blindness by 90.20% of validation accuracy. The freezing and unfreezing for fine-tuned EfficientNet-B5 significantly improved the prediction with 92.32% validation accuracy. The proposed approach has been developed and tested only for the early detection of diabetic retinopathy in diabetic patients. We also performed oversampling strategies for the interpretation on our detected results. For other medical image diagnosis, the approach needs to be tested before making any concluding remarks. We found that there are more number of 0's and 2's i.e., no diabetic retinopathy symptom and moderate retinopathy shown in around 89% of the images.

We intend to adopt other CNN architectures [43]–[45] such as UNet with ResNet models and EfficientNet weights [46] with UNet for such imbalanced image collection, and pseudo labeling the imbalanced dataset may potentially improve the prediction for given classes including consideration of this identification task as binary classification by individually labeling the data could be an added advantage. The limitation in doing so is of processing power which greatly increases if using EfficientNet-B6 or B7 weights. The proposed approach has been developed and tested only for the early detection of diabetic retinopathy in diabetic patients.

## DECLARATION OF COMPETING INTEREST

We wish to confirm that there are no known conflicts of interest associated with this publication.

## REFERENCES

[1] M. S. Hossain and G. Muhammad, "Deep learning based pathology detection for smart connected healthcare," *IEEE Netw.*, vol. 34, no. 6, pp. 120–125, Nov. 2020.

[2] M. Alhussein, G. Muhammad, M. S. Hossain, and S. U. Amin, "Cognitive IoT-cloud integration for smart healthcare: Case study for epileptic seizure detection and monitoring," *Mobile Netw. Appl.*, vol. 23, no. 6, pp. 1624–1635, Dec. 2018.

[3] J. Qian, P. Tiwari. S. P. Gochhayat, and H. M. Pandey, "A noble double-dictionary-based ECG compression technique for IoTH," *IEEE Internet Things J.*, vol. 7, no. 10, pp. 10160–10170, Feb. 2020.

[4] M. S. Hossain, G. Muhammad, and A. Alamri, "Smart healthcare monitoring: A voice pathology detection paradigm for smart cities," *Multimedia Syst.*, vol. 25, no. 5, pp. 565–575, Oct. 2019.

[5] W. H. Organization. (2016). *Death Among Diabetic Patients*. [Online]. Available: http://www.who.int/diabetes/global-report/en

[6] P. Feng, Y. Pan, B. Wei, W. Jin, and D. Mi, "Enhancing retinal image by the contourlet transform," *Pattern Recognit. Lett.*, vol. 28, no. 4, pp. 516–522, Mar. 2007.

[7] Z. Liu, C. Opas, and S. M. Krishnan, "Automatic image analysis of fundus photograph," in *Proc. 19th Annu. Int. Conf. Eng. Med. Biol. Soc. Magnificent Milestones Emerg.*, 1997, pp. 524–525.

[8] L. M. Ruta, D. J. Magliano, R. LeMesurier, H. R. Taylor, P. Z. Zimmet, and J. E. Shaw, "Prevalence of diabetic retinopathy in type 2 diabetes in developing and developed countries," *Diabetic Med.*, vol. 30, no. 4, pp. 387–398, Apr. 2013.
[9] IDF Team. (2009). *Latest Diabetes Figures Paint Grim Global Picture*. [Online]. Available: http://www.idf.org/press_releases
[10] S. U. Amin, M. Alsulaiman, G. Muhammad, M. A. Mekhtiche, and M. S. Hossain, "Deep learning for EEG motor imagery classification based on multi-layer CNNs feature fusion," *Future Gener. Comput. Syst.*, vol. 101, pp. 542–554, Dec. 2019.
[11] S. M. Anwar, M. Majid, A. Qayyum, M. Awais, and M. Alnowami, "Medical image analysis using convolutional neural networks: A review," *J. Med. Syst.*, vol. 42, p. 226, Oct. 2018.
[12] X. Yang, T. Zhang, C. Xu, S. Yan, M. S. Hossain, and A. Ghoneim, "Deep relative attributes," *IEEE Trans. Multimedia*, vol. 18, no. 9, pp. 1832–1842, Sep. 2016.
[13] N. Tajbakhsh, J. Y. Shin, S. R. Gurudu, R. T. Hurst, C. B. Kendall, M. B. Gotway, and J. Liang, "Convolutional neural networks for medical image analysis: Full training or fine tuning?" *IEEE Trans. Med. Imag.*, vol. 35, no. 5, pp. 1299–1312, May 2016.
[14] G. Muhammad, M. S. Hossain, and N. Kumar, "EEG-based pathology detection for home health monitoring," *IEEE J. Sel. Areas Commun.*, vol. 39, no. 2, pp. 603–610, Feb. 2021.
[15] M. S. Hossain, G. Muhammad, and N. Guizani, "Explainable AI and mass surveillance system-based healthcare framework to combat COVID-I9 like pandemics," *IEEE Netw.*, vol. 34, no. 4, pp. 126–132, Jul. 2020.
[16] W. Tan, P. Tiwari, H. M. Pandey, C. Moreira, and A. K. Jaiswal, "Multimodal medical image fusion algorithm in the era of big data," *Neural Comput. Appl.*, pp. 1–21, Jul. 2020.
[17] M. Tan and Q. V. Le, "EfficientNet: Rethinking model scaling for convolutional neural networks," 2019, *arXiv:1905.11946*. [Online]. Available: http://arxiv.org/abs/1905.11946
[18] N. Silberman, K. Ahrlich, R. Fergus, and L. Subramanian, "Case for automated detection of diabetic retinopathy," in *Proc. AAAI Spring Symp.*, 2010.
[19] M. Chen, J. Yang, L. Hu, M. S. Hossain, and G. Muhammad, "Urban healthcare big data system based on crowdsourced and cloud-based air quality indicators," *IEEE Commun. Mag.*, vol. 56, no. 11, pp. 14–20, Nov. 2018.
[20] S. Muhammad Saiful Islam, M. Mahedi Hasan, and S. Abdullah, "Deep learning based early detection and grading of diabetic retinopathy using retinal fundus images," 2018, *arXiv:1812.10595*. [Online]. Available: http://arxiv.org/abs/1812.10595
[21] P. Tiwari, S. Uprety, S. Dehdashti, and M. S. Hossain, "Terminformer: Unsupervised term mining and analysis in biomedical literature," in *Proc. Neural Comput. Appl.*, Sep. 2020, pp. 1–14.
[22] M. S. Hossain and G. Muhammad, "Emotion-aware connected healthcare big data towards 5G," *IEEE Internet Things J.*, vol. 5, no. 4, pp. 2399–2406, Aug. 2018.
[23] S. Wan, Y. Liang, and Y. Zhang, "Deep convolutional neural networks for diabetic retinopathy detection by image classification," *Comput. Electr. Eng.*, vol. 72, pp. 274–282, Nov. 2018.
[24] L. Hu, M. Qiu, J. Song, M. S. Hossain, and A. Ghoneim, "Software defined healthcare networks," *IEEE Wireless Commun.*, vol. 22, no. 6, pp. 67–75, Dec. 2015.
[25] M. S. Hossain, "Cloud-supported cyber–physical localization framework for patients monitoring," *IEEE Syst. J.*, vol. 11, no. 1, pp. 118–127, Mar. 2017.
[26] A. Kumar and P. Vashist, "Indian community eye care in 2020," *Indian J. Ophthalmol.*, vol. 68, pp. 291–293, Oct. 2020.
[27] S. Bhalla, T. Soni, M. Joshi, V. Sharma, R. Mishra, V. Mohan, R. Unnikrishnan, R. Kim, G. S. Murthy, D. Prabhakaran, P. Rani, and R. Rajalakshmi, "A partnership model for capacity-building of primary care physicians in evidence-based management of diabetic retinopathy in india," *Indian J. Ophthalmol.*, vol. 68, no. 13, p. 67, 2020.
[28] A. K. Jaiswal, P. Tiwari, S. Garg, and M. S. Hossain, "Entity-aware capsule network for multi-class classification of big data: A deep learning approach," *Future Gener. Comput. Syst.*, vol. 117, pp. 1–11, Apr. 2021.
[29] S. U. Amin, M. S. Hossain, G. Muhammad, M. Alhussein, and M. A. Rahman, "Cognitive smart healthcare for pathology detection and monitoring," *IEEE Access*, vol. 7, pp. 10745–10753, 2019.
[30] P. Costa, T. Araujo, G. Aresta, A. Galdran, A. M. Mendonca, A. Smailagic, and A. Campilho, "EyeWeS: Weakly supervised pre-trained convolutional neural networks for diabetic retinopathy detection," in *Proc. 16th Int. Conf. Mach. Vis. Appl. (MVA)*, May 2019, pp. 1–6.
[31] W. Min, B.-K. Bao, C. Xu, and M. S. Hossain, "Cross-platform multimodal topic modeling for personalized inter-platform recommendation," *IEEE Trans. Multimedia*, vol. 17, no. 10, pp. 1787–1801, Oct. 2015.
[32] Z. Wang and J. Yang, "Diabetic retinopathy detection via deep convolutional networks for discriminative localization and visual explanation," 2017, *arXiv:1703.10757*. [Online]. Available: http://arxiv.org/abs/1703.10757
[33] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. pattern Recognit.*, 2016, pp. 770–778.
[34] J. Cohen, "Weighted kappa: Nominal scale agreement provision for scaled disagreement or partial credit," *Psychol. Bull.*, vol. 70, no. 4, pp. 213–220, 1968.
[35] M. J. Warrens, "Some paradoxical results for the quadratically weighted kappa," *Psychometrika*, vol. 77, no. 2, pp. 315–323, Apr. 2012.
[36] A. D'Addabbo and R. Maglietta, "Parallel selective sampling method for imbalanced and large data classification," *Pattern Recognit. Lett.*, vol. 62, pp. 61–67, Sep. 2015.
[37] A. Buslaev, A. Parinov, E. Khvedchenya, V. I. Iglovikov, and A. A. Kalinin, "Albumentations: Fast and flexible image augmentations," 2018, *arXiv:1809.06839*. [Online]. Available: http://arxiv.org/abs/1809.06839
[38] G. Vandewiele, I. Dehaene, G. Kovács, L. Sterckx, O. Janssens, F. Ongenae, F. De Backere, F. De Turck, K. Roelens, J. Decruyenaere, S. Van Hoecke, and T. Demeester, "Overly optimistic prediction results on imbalanced data: A case study of flaws and benefits when applying oversampling," *Artif. Intell. Med.*, vol. 111, Jan. 2021, Art. no. 101987.
[39] Y. N. D. D. L.-P. Hongyi Zhang and M. Cisse, "Mixup: Beyond empirical risk minimization," *Int. Conf. Learn. Represent.*, 2018, pp. 1–13. [Online]. Available: https://openreview.net/forum?id=r1Ddp1-Rb
[40] M. S. Ayhan and P. Berens, "Test-time data augmentation for estimation of heteroscedastic aleatoric uncertainty in deep neural networks," in *Proc. Int. Conf. Med. Imag. Deep Learn.*, 2018.
[41] S. Singh, N. K. Rajput, V. K. Rathi, H. M. Pandey, A. K. Jaiswal, and P. Tiwari, "Securing blockchain transactions using quantum teleportation and quantum digital signature," *Neural Process. Lett.*, pp. 1–16, Jun. 2020.
[42] S. Kumar, P. Tiwari, and M. Zymbler, "Internet of Things is a revolutionary approach for future technology enhancement: A review," *J. Big Data*, vol. 6, no. 1, pp. 1–21, Dec. 2019.
[43] A. K. Jaiswal, P. Tiwari, S. Kumar, D. Gupta, A. Khanna, and J. J. P. C. Rodrigues, "Identifying pneumonia in chest X-rays: A deep learning approach," *Measurement*, vol. 145, pp. 511–518, Oct. 2019.
[44] A. Kumar Jaiswal, I. Panshin, D. Shulkin, N. Aneja, and S. Abramov, "Semi-supervised learning for cancer detection of lymph node metastases," 2019, *arXiv:1906.09587*. [Online]. Available: http://arxiv.org/abs/1906.09587
[45] D. Nandini, E. Capecci, L. Koefoed, I. La na, G. K. Shahi, and N. Kasabov, "Modelling and analysis of temporal gene expression data using spiking neural networks," in *Proc. Int. Conf. Neural Inf. Process.* Cham, Switzerland: Springer, 2018, pp. 571–581.
[46] A. K. Jaiswal, P. Tiwari, V. K. Rathi, J. Qian, H. M. Pandey, and V. H. C. Albuquerque, "Covidpen: A novel COVID-19 detection model using chest X-rays and CT scans," *Medrxiv*, Jan. 2020.
[47] A. K. Sangaiah, D. V. Medhane, G. Bian, A. Ghoneim, M. Alrashoud, and M. S. Hossain, "Energy-aware green adversary model for cyberphysical security in industrial system," *IEEE Trans. Ind. Informat.*, vol. 16, no. 5, pp. 3322–3329, May 2020.



**AMIT KUMAR JAISWAL** is currently pursuing the Ph.D. degree with the University of Bedfordshire. He was working as a Marie Curie Researcher in quantum information retrieval with the University of Bedfordshire. He is also a Research Software Engineer with the School of Mathematics, University of Leeds. His research interests include existing quantum-IR frameworks, applied machine learning, and their applications in Web search, medical imaging, and financial domain in general.



**PRAYAG TIWARI** received the M.S. degree from NUST MISIS, Moscow. He is currently pursuing the Ph.D. degree with the University of Padova, Italy. He is currently working as a Postdoctoral Researcher with Aalto University. Previously, he was working as a Marie Curie Researcher with the University of Padova. He is also worked as a Research Assistant with NUST MISIS, where he has had teaching and industrial work experience. He has several publications in top journals and conferences. His research interests include machine learning, deep learning, quantum-inspired machine learning, information retrieval, health informatics, and the IoT.



**SACHIN KUMAR** (Senior Member, IEEE) received the Ph.D. degree in data mining/machine learning from the Indian Institute of Technology Roorkee, in 2017. He is currently associated with the Department of Computer Science, South Ural State University, Chelyabinsk, Russia. His research interests include intelligent transportation systems, machine learning, data mining, and health informatics.



**MABROOK S. AL-RAKHAMI** (Member, IEEE) received the master's degree in information systems from King Saud University, Saudi Arabia, where he is currently pursuing the Ph.D. degree with the College of Computer and Information Sciences. He has authored several articles in peer-reviewed IEEE/ACM/Springer/Wiley journals. His research interests include edge intelligence, cloud computing, the IoT, and big data.



**MUBARAK ALRASHOUD** (Member, IEEE) received the Ph.D. degree in computer science from Ryerson University, Toronto, ON, Canada, in 2015. He is currently an Associate Professor and the Head of the Department of Software Engineering, College of Computer and Information Sciences, King Saud University, Riyadh, Saudi Arabia.



**AHMED GHONEIM** (Senior Member, IEEE) received the Ph.D. degree in software engineering from the University of Magdeburg, Germany, in 2007. He is currently an Associate Professor with the Department of Software Engineering, College of Computer and Information Sciences, King Saud University, Riyadh, Saudi Arabia. His research interests include software evolution, service-oriented engineering, software development methodologies, net-centric computing, human–computer interaction, cloud networking, smart environment, such as smart city and smart health, AI, deep learning, edge computing, and the Internet of Things (IoT).